Blessed G/Michael Catholic Secondary School

Semester 1 Aptitude Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** What number comes next in the geometric pattern: 3, 6, 18, 72, 360, ...?
A. 1080 **B.** 1440 **C.** 1800 **D.** 2160

____ **2.** Find the missing term in the sequence: 4, 9, 25, 49, 121, 169, ...?
A. 196 **B.** 225 **C.** 289 **D.** 361

____ **3.** A shopkeeper buys an article for 400 Birr and sells it for 500 Birr. What is the percentage profit?
A. 20% **B.** 25% **C.** 30% **D.** 15%

____ **4.** If 12 men can complete a construction ditch in 8 days, how many days will 6 men take working at the identical rate?
A. 4 days **B.** 12 days **C.** 16 days **D.** 24 days

____ **5.** The average (arithmetic mean) of five numbers is 24. If one number is removed, the average of the remaining four numbers becomes 22. What number was removed?
A. 28 **B.** 30 **C.** 32 **D.** 34

____ **6.** A car travels from City A to City B at an average speed of 60 km/h and returns at 40 km/h. What is the average speed for the round trip?
A. 50 km/h **B.** 48 km/h **C.** 52 km/h **D.** 45 km/h

____ **7.** The ratio of the ages of a father and son is 7 : 3. If the sum of their ages is 60 years, how old is the son?
A. 14 years **B.** 18 years **C.** 21 years **D.** 24 years

____ **8.** A bag contains 5 red balls, 4 green balls, and 3 blue balls. If one ball is drawn at random, what is the probability that it is NOT blue?
A. 1/4 **B.** 3/4 **C.** 5/12 **D.** 7/12

____ **9.** If 20% of a number is equal to 45, what is 80% of that same number?
A. 90 **B.** 135 **C.** 180 **D.** 225

____ **10.** A pipe can fill a reservoir in 4 hours, while a drain pipe can empty it in 6 hours. If both pipes are opened simultaneously, how many hours will it take to fill the reservoir?
A. 10 hours **B.** 12 hours **C.** 14 hours **D.** 24 hours

____ **11.** Complete the analogy: TELESCOPE : ASTRONOMER :: MICROSCOPE : _____
A. CHEMIST **B.** BIOLOGIST **C.** GEOLOGIST **D.** ARCHAEOLOGIST

____ **12.** Select the pair that displays the same relationship: CULPABLE : BLAME ::
A. Innocent : Punishment **B.** Laudable : Praise
C. Vulnerable : Protection **D.** Careless : Success

____ **13.** Complete the analogy: THERMOMETER : TEMPERATURE :: BAROMETER : _____
A. HUMIDITY **B.** AIR PRESSURE **C.** WIND SPEED **D.** PRECIPITATION

____ **14.** Choose the word most nearly OPPOSITE in meaning to EPHEMERAL:
A. Transient **B.** Permanent **C.** Fleeting **D.** Fragile

____ **15.** Complete the analogy: DROUGHT : RAINFALL :: FAMINE : _____
A. POVERTY **B.** FOOD **C.** HARVEST **D.** DESERT

____ **16.** Select the word that does NOT belong with the others:
A. Triangle **B.** Hexagon **C.** Circle **D.** Pentagon

____ **17.** Complete the analogy: AUTHOR : NOVEL :: COMPOSER : _____

A. SYMPHONY **B.** INSTRUMENT **C.** ORCHESTRA **D.** LYRICS

____ **18.** Premise 1: All mammals breathe oxygen.
Premise 2: All dolphins are mammals.
Conclusion: Therefore, all dolphins breathe oxygen.
This deductive argument is:

A. Valid and sound **B.** Invalid but sound

C. Valid but unsound **D.** Fallacious deduction

____ **19.** Premise 1: Some athletes are musicians.
Premise 2: All musicians love art.
Which conclusion logically follows?

A. All athletes love art **B.** Some athletes love art

C. No athlete loves art **D.** All musicians are athletes

____ **20.** If 'No reptiles have feathers' is a TRUE statement, which of the following statements must be FALSE?

A. All reptiles have feathers **B.** Some reptiles have feathers

C. No feathers belong to reptiles **D.** Both A and B

____ **21.** In a code language, if BRAIN is coded as 'CSBJO', how is LOGIC coded in the same system?

A. MPHKD **B.** MQHJD **C.** NPHKE **D.** MPHJE

____ **22.** Pointing to a photograph of a man, Sara said: 'His mother is the only daughter of my mother.' How is Sara related to the man?

A. Sister **B.** Aunt **C.** Mother **D.** Grandmother

____ **23.** Five students (Abebe, Bekele, Chala, Dawit, Ephrem) sit in a straight row facing north. Bekele is to the immediate right of Abebe. Dawit is between Chala and Ephrem. Chala is at the extreme left. Who sits in the middle?

A. Bekele **B.** Dawit **C.** Abebe **D.** Ephrem

____ **24.** If + means multiply, - means divide, × means add, and ÷ means subtract, what is the value of: 10 + 5 × 12 - 4 ÷ 8?

A. 45 **B.** 53 **C.** 55 **D.** 60

____ **25.** A clock shows 3:00 o'clock. What is the angle in degrees between the hour hand and the minute hand?

A. 45° **B.** 60° **C.** 90° **D.** 120°

____ **26.** If today is Tuesday, what day of the week will it be exactly 100 days from today?

A. Wednesday **B.** Thursday **C.** Friday **D.** Saturday

____ **27.** Complete the analogy: EXPAND : CONTRACT :: ACCELERATE : _____

A. DECELERATE **B.** MULTIPLY **C.** MIGRATE **D.** ROTATE

____ **28.** A man walks 6 km North, then turns East and walks 8 km. How far is he from his starting point in a straight line?

A. 10 km **B.** 12 km **C.** 14 km **D.** 16 km

____ **29.** In a class of 50 students, 30 play Football, 25 play Basketball, and 10 play both sports. How many students play NEITHER sport?

A. 5 **B.** 10 **C.** 15 **D.** 20

____ **30.** Find the odd one out among the following numbers: 27, 64, 125, 144, 216

A. 64 **B.** 125 **C.** 144 **D.** 216